



V-Guard Sentinel

The inverter already knows when the power goes. Sentinel teaches it what to keep on — and when its battery will quit.

Track 4 · From Smart Products to Intelligent Products

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Finale · Kochi · 24 September 2026



10:03 pm. The battery dies — with no warning.

Three hours into a power cut, Home 6 in our village goes dark. Home 1 has the same battery and appliances — and stays lit.



85 %

of Indian households face a power cut every day (LocalCircles, 2023)

×2 wear

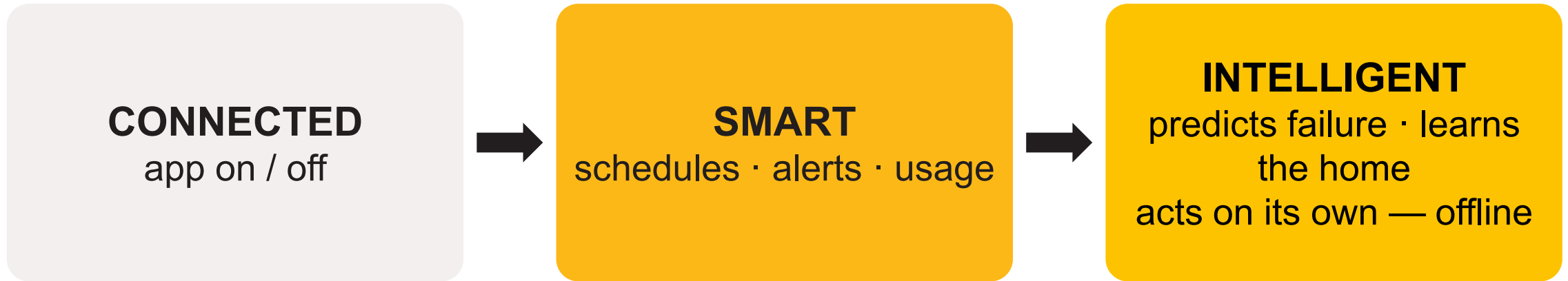
every +10 °C: lead-acid life halves with heat — the Indian summer is the battery's enemy

≈ 1.5 %

of V-Guard revenue goes to warranty (≈ ₹69 cr, FY24)

Sentinel turns a smart inverter into an intelligent one.

Smart products take commands. Intelligent products predict, decide and act — without the cloud.

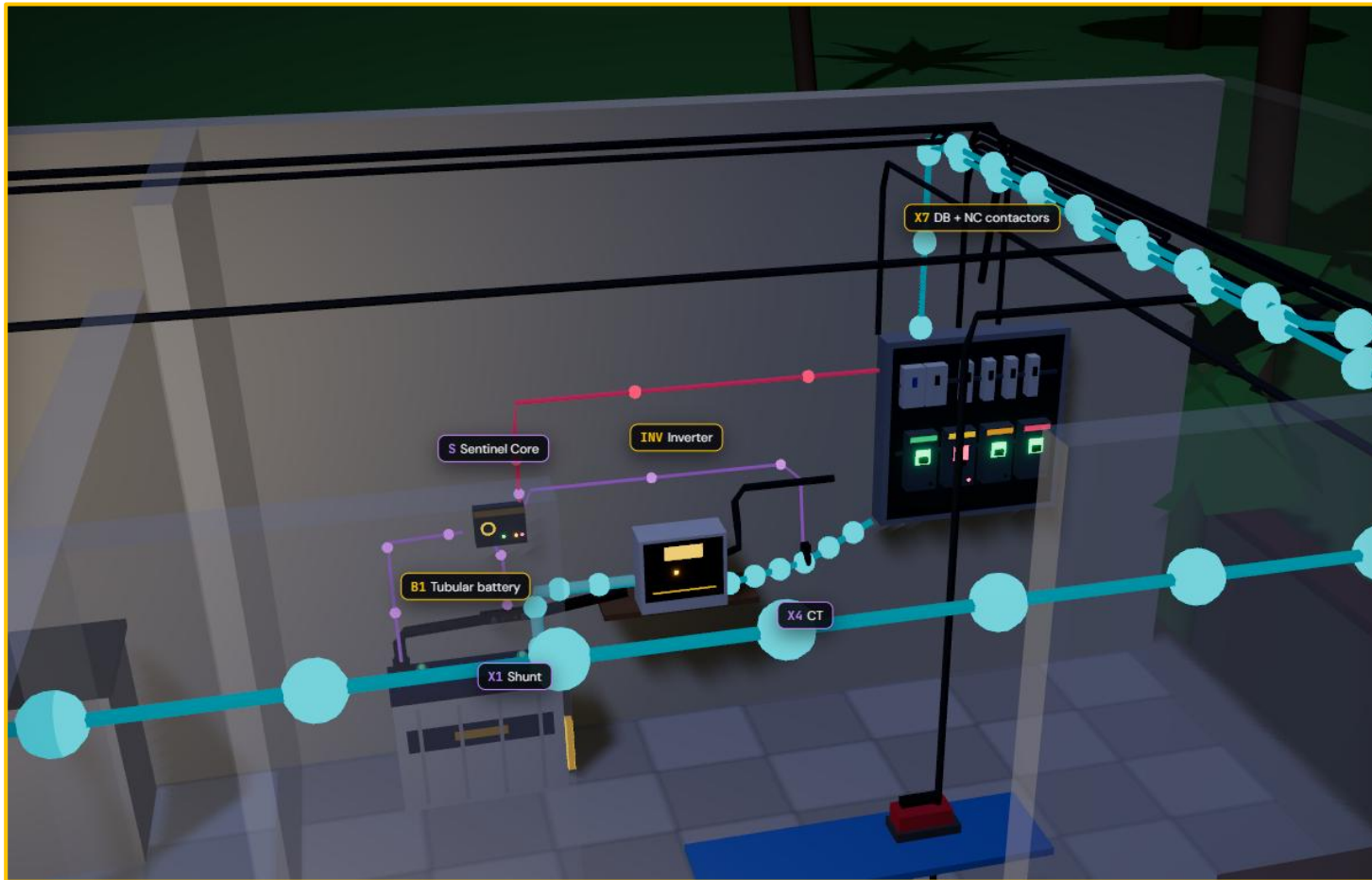


← **Sentinel**

- 1 Embedded** a daughter-board inside new V-Guard inverters
- 2 Retrofit** a 30-minute electrician fit on the installed base
- 3 Offline-first** every core function works with zero internet

One small box reads the battery and switches the loads.

Nothing on the battery or the inverter is modified on the retrofit — only the negative cable passes through a shunt.



- 1 Shunt in the battery – lead**
battery current → INA228
- 2 NTC on the battery post**
temperature: the #1 ageing driver
- 3 CT + voltage tap on AC-OUT**
real & reactive power → ATM90E32AS
- 4 4 NC contactors in the DB**
coil off = load ON, always
- 5 ESP32-S3 Sentinel Core**
90 × 70 × 35 mm, powered by the battery

Four jobs, one board — all on the ESP32, no cloud needed.

1

Battery health

Predicts a replacement window — 'replace in 6–14 weeks'

2

Habit autopilot

Keeps the fridge, lights and router alive during a cut

3

Energy coach + grid shield

Names the big appliances; logs every sag and swell

4

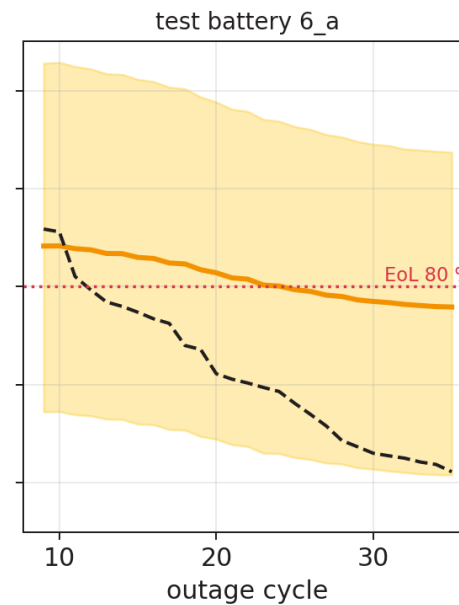
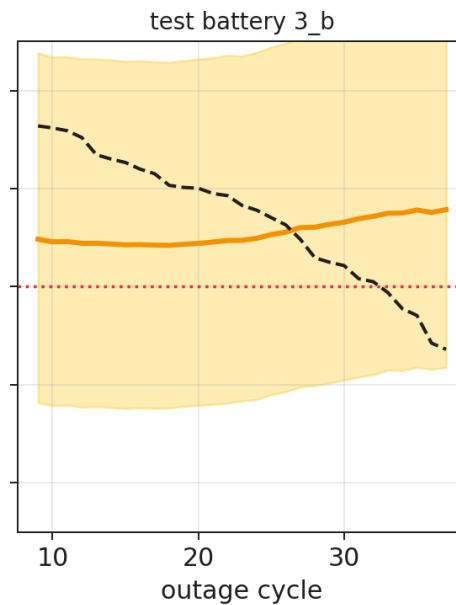
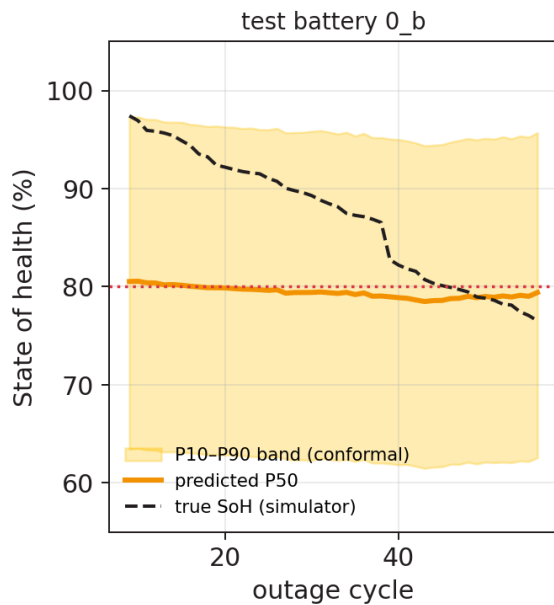
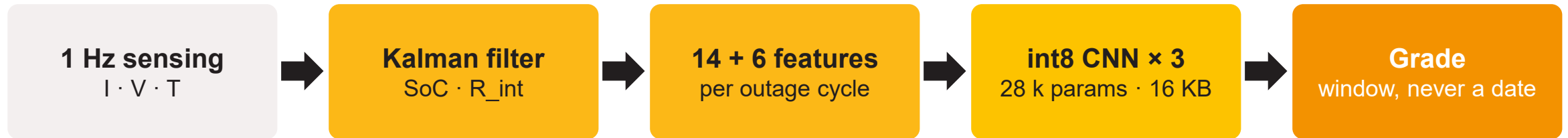
Signed health log

Tamper-proof record — warranty decided by evidence

Same core, same sensors, different models → the platform extends to pumps, stabilisers and water heaters.

It gives a window, not a date: "replace in 6–14 weeks".

Physics first, learning second — every step runs on the ESP32.



8.2 pt

SoH error on 3 never-seen test batteries (synthetic data — honest, not flattering)

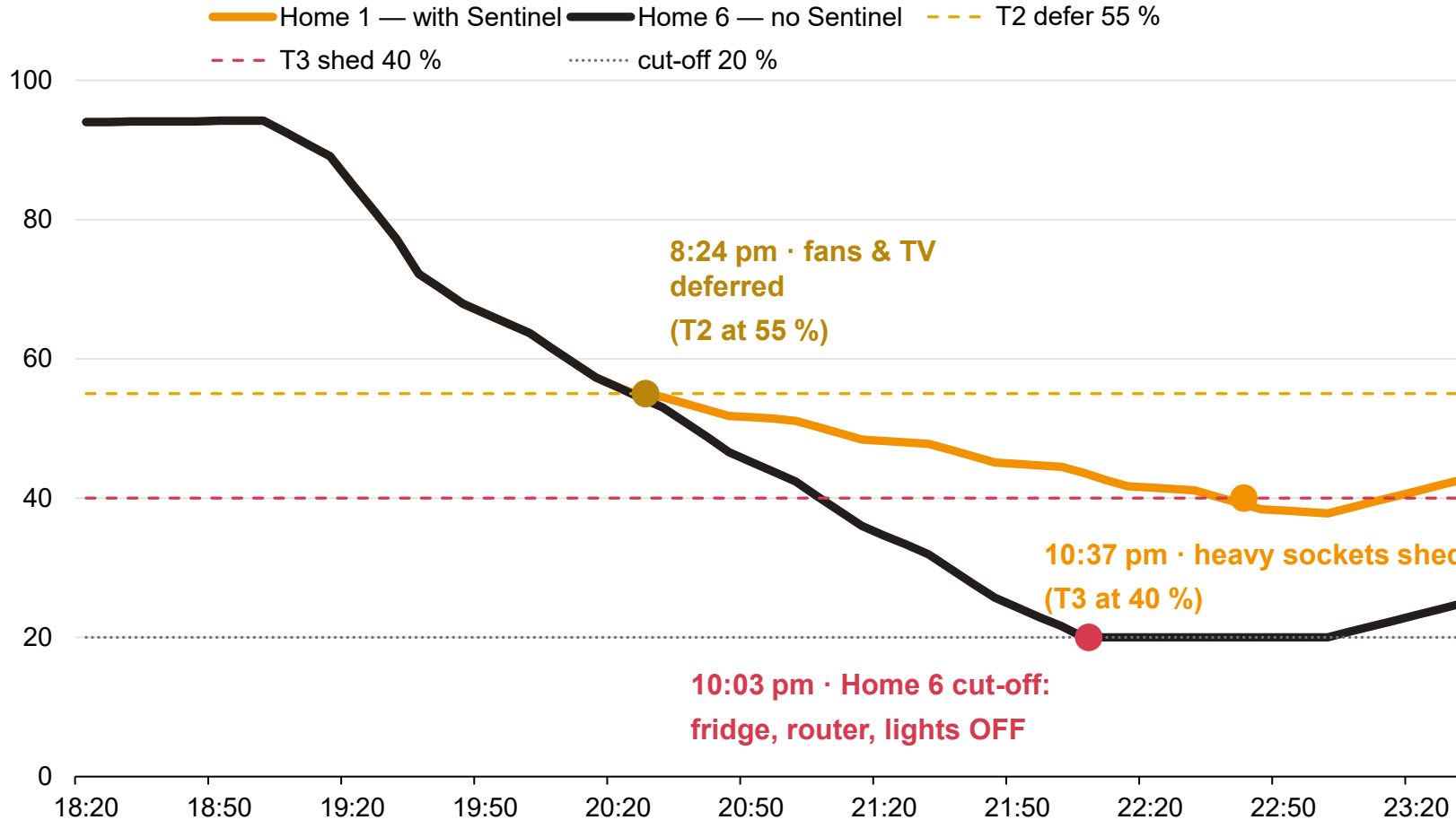
+0.01 pt

accuracy lost going float → int8 — it fits on the chip for free

Band is calibrated but wide today; the Kochi aging campaign narrows it.

Same outage, same battery: only Home 1 keeps its fridge on.

Battery charge through the 7–11 pm cut, straight from our simulation engine; thresholds from design doc 05.



57 min

Home 6 sat in the dark until mains returned at 11 pm

0 min

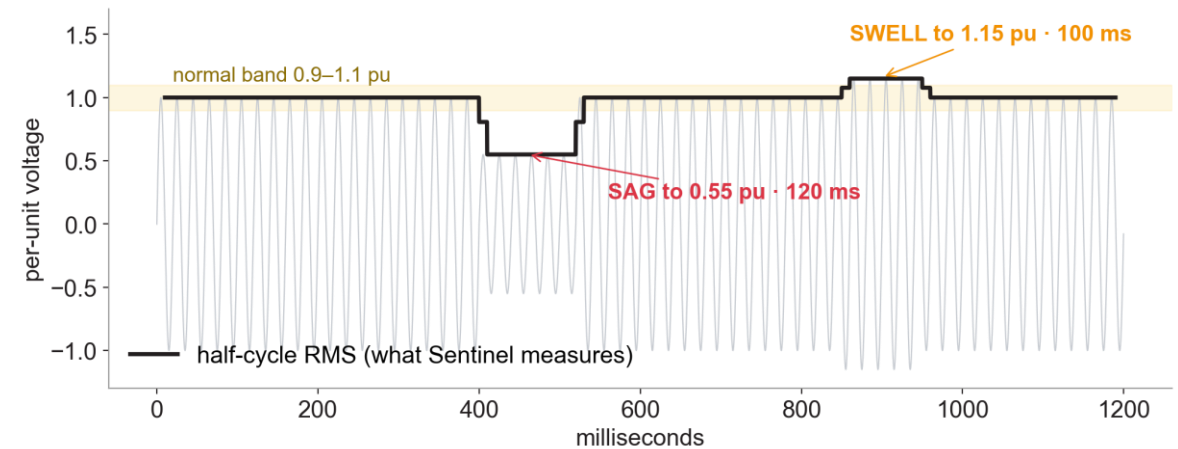
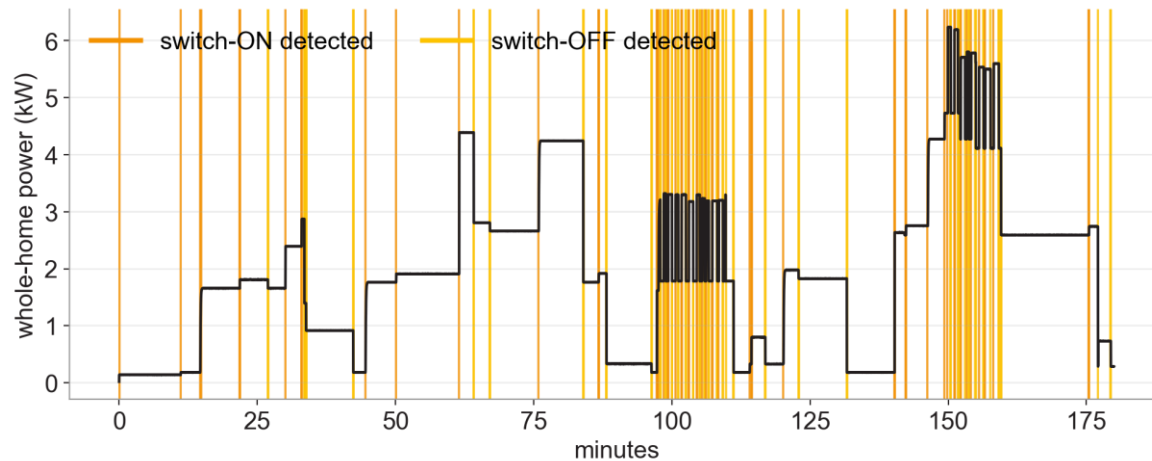
Home 1's fridge, router and hall light were never off

Coil off = load on.

NC contactors, a hardware supervisory timer and a medical jumper: any crash or reset brings every load back.

One sensor saves energy and catches a misbehaving grid.

Both run on the metering front-end (ATM90E32AS) and the isolated voltage tap (AMC1311) — outputs of the team's own detector code.

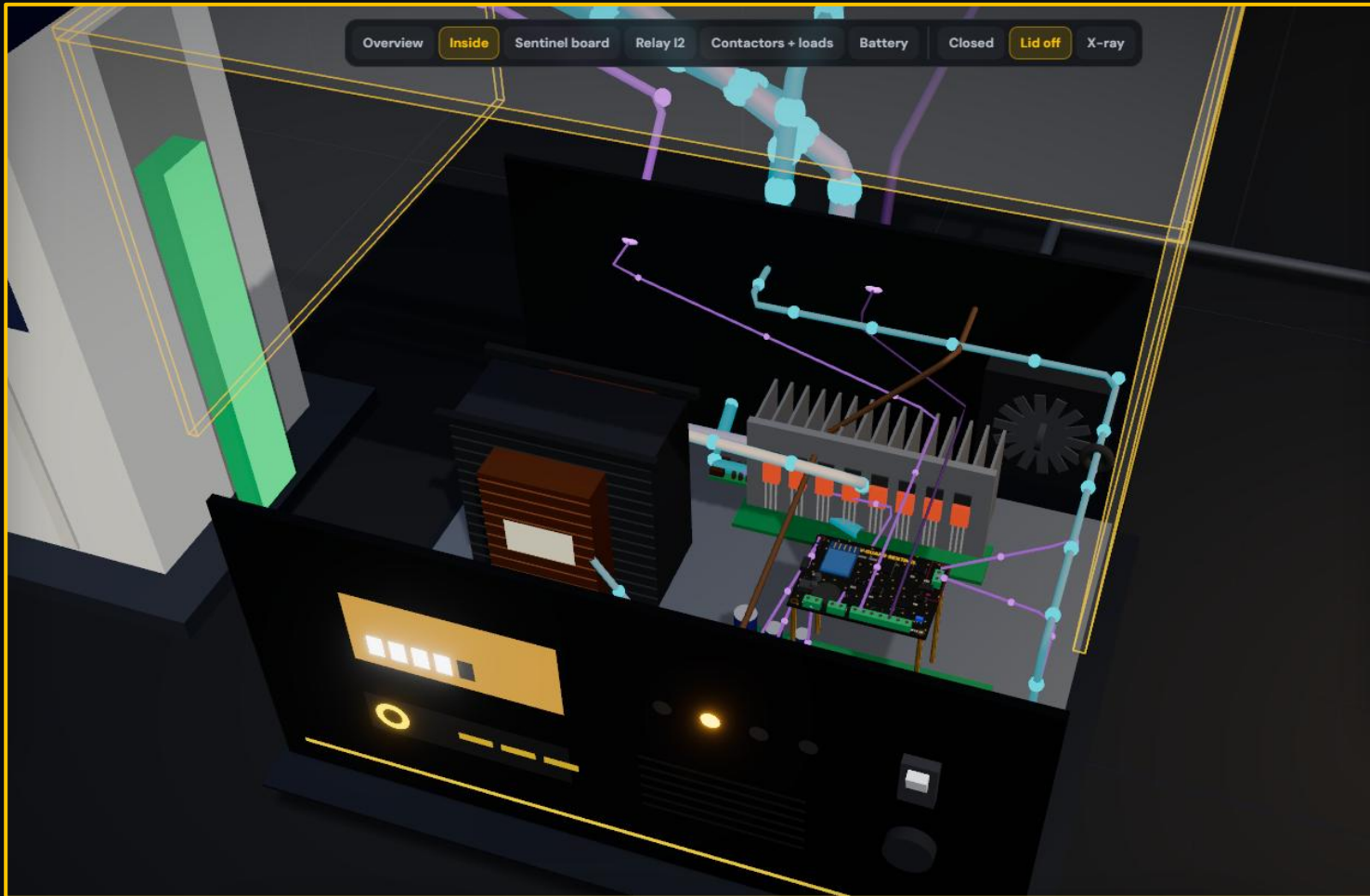


Energy Coach 123 switch-ON/OFF events found in 3 h of whole-home power; each becomes a 13-number signature. Fridge F1 0.97, iron 1.00.

Grid Shield a 120 ms sag to 0.55 pu measured as 0.54; a 100 ms swell to 1.15 pu as 1.16. 13 / 13 dips caught. Logged, dated, signed.

Inside the inverter, Sentinel taps what's already there.

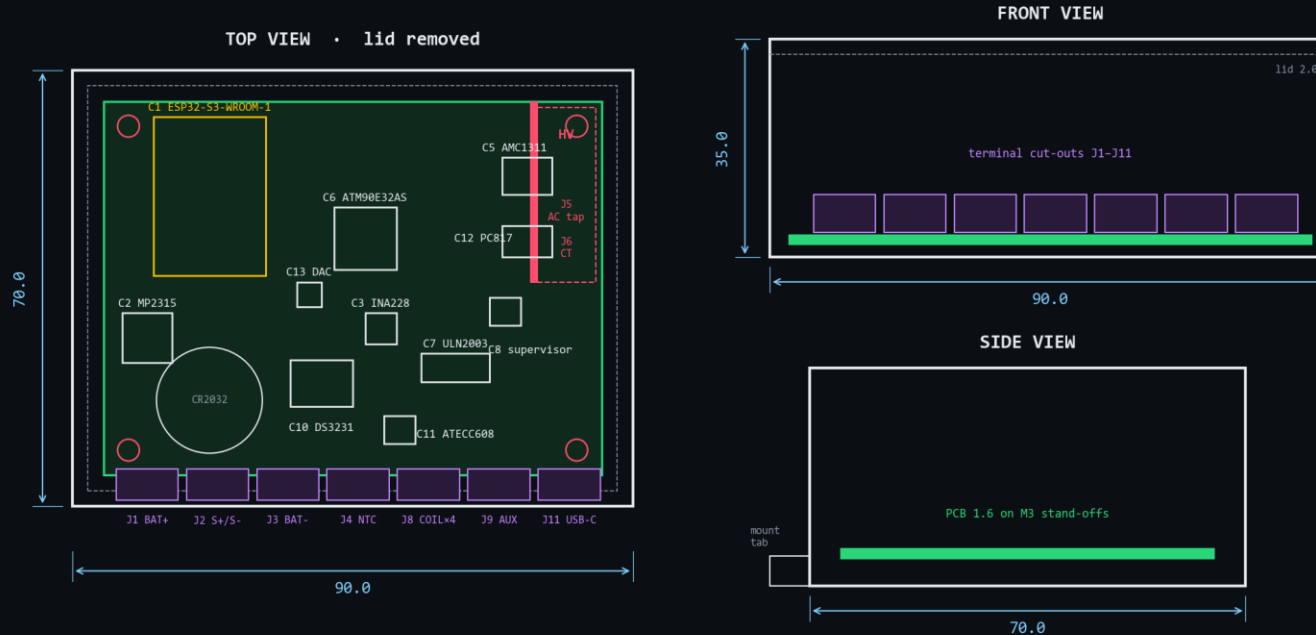
Our 3D proof of concept of the Embedded SKU: lid off, every part clickable, live power flow.



- 1 Inverter's own shunt (I8)**
→ INA228 · battery current
- 2 Mode / mains line**
→ PC817 opto · outage vote s2
- 3 SG3525 charge feedback**
→ MCP4725 DAC · temperature-compensated charging
- 4 AC-OUT rail**
→ AMC1311 + ATM90E32AS · V, P, Q
- 5 4 coil outputs**
→ NC contactor panel · T1 / T2 / T3 / medical

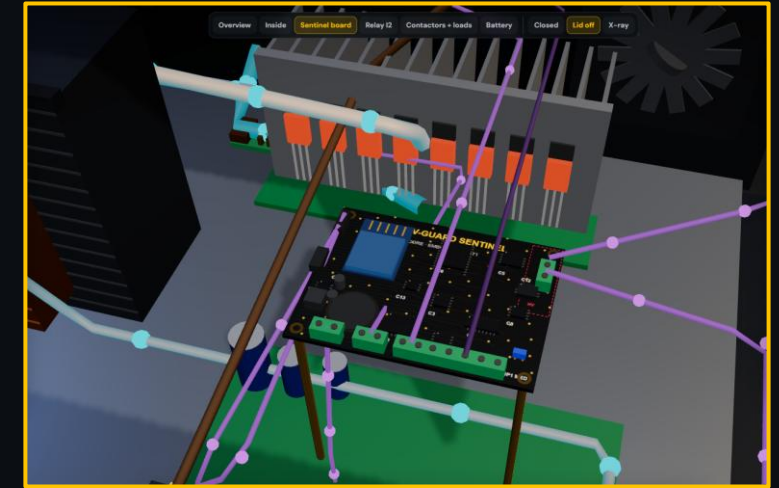
Designed to be built: 90 × 70 × 35 mm, HV zone fenced off.

Mechanical drawing from our round-3 spec, and the same board as a 3D render in the simulation.



V-GUARD SENTINEL CORE – ENCLOSURE & PCB

Enclosure 90 × 70 × 35 mm · PC/ABS UL94 V-0 · wall 2.5 · IP20
PCB 80 × 60 × 1.6 · 4× M3 (Ø3.5) at 4 mm inset
Battery current never enters the PCB – only shunt mV (J2)
AMC1311 + PC817 straddle the HV isolation moat
Coil off = load on: ULN2003 + supervisory timer + JP1
Units mm · per prototype/02-03 (round 3) · Rev T1



Battery current never enters the PCB
only the shunt's millivolts do.

Isolation moat

AMC1311 and PC817 straddle the HV zone.

UL94 V-0 PC/ABS, 4× M3

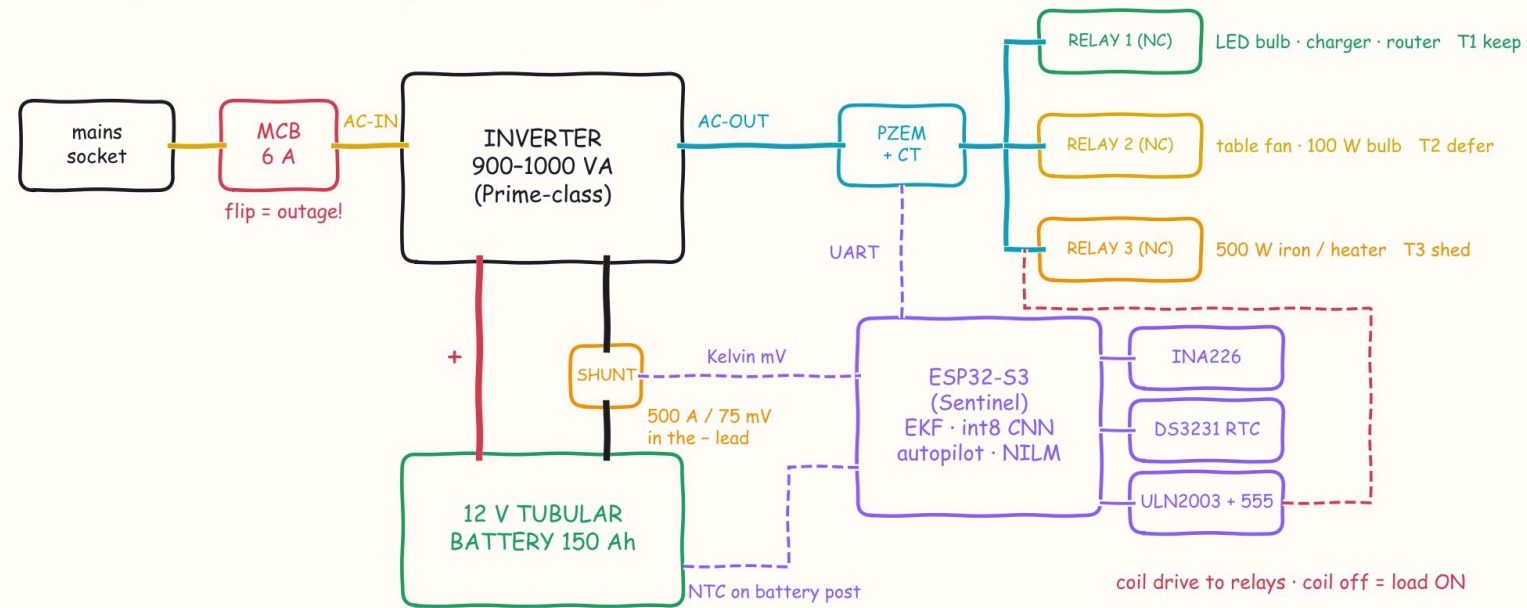
DIN or screw mount beside the inverter.

100%

Tier-0 demonstrator from prototype doc 05 — every part available from Indian distributors today.



real battery · real inverter · three relay-switched tiers · nothing on the internet



Reset the ESP32 while T3 is shed → relay 3 re-closes in about a second.

Let's watch it happen — live.

Sentinel Live runs offline in one HTML file. Three moments to watch for:



1 · The village

Feeder cuts at 7 pm. Watch Home 6 go dark at 10:03 while its twin stays lit.



2 · Flip the MCB

Relay swings in slow motion: 8 ms gap on the scope, outage confirmed in 1.5 s.



3 · Crash it

Drop SoC below 40 % → the iron sheds. Reset the chip → every load comes back.

What's proven, what's synthetic, and what still needs hardware.

We'd rather you trust three honest columns than one shiny number.

245

automated tests, all passing

6

C modules bit-identical to Python

3 %

Kalman SoC error vs 7.2 %/week
naive

13 / 13

grid dips detected

PROVEN IN CODE

- int8 model exported + golden self-test in C
- autopilot: 50 / 50 C scenarios, zero chatter
- signed log: tamper, delete, replay caught

SYNTHETIC ONLY

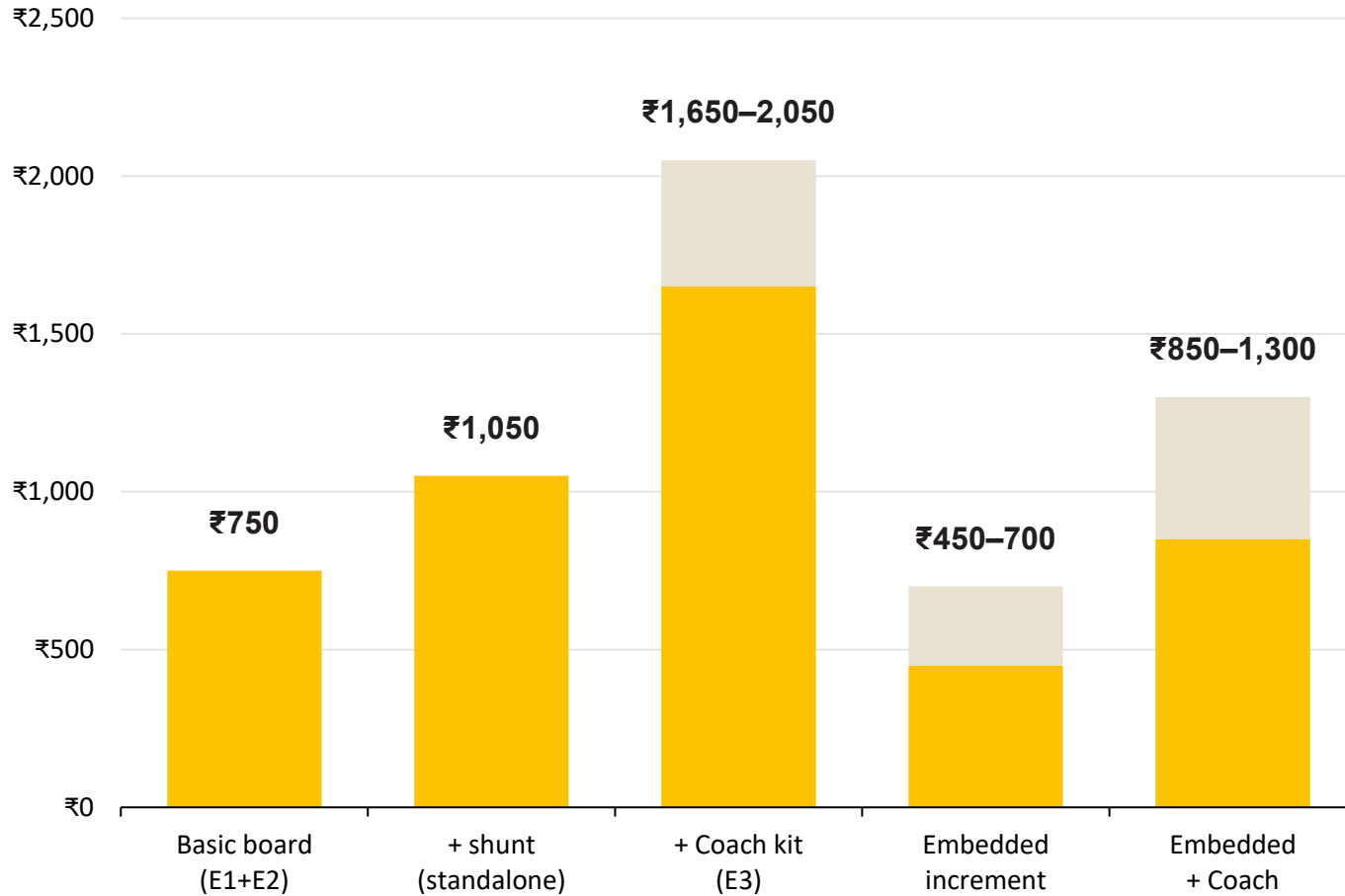
- SoH 8.2 pt on simulated tubular batteries
- RUL band calibrated (0.99) but wide
- NILM / PQ on synthetic streams

NEEDS HARDWARE / DATA

- ESP32 flash + latency on silicon
- real bench: battery, CT, relays
- aging campaign → real SoH accuracy

Two SKUs, one board — for 5–14 % of what one battery costs.

Cost stated by the engines each tier enables (design decision D7), against a ₹15 k tubular battery.



Sentinel-Embedded

Inside new V-Guard inverters. Taps the inverter's own shunt; controls the charger through a DAC with a hardware ceiling. Life-extension claims live here only.

Sentinel-Retrofit

Box beside any inverter. External shunt, NTC, CT and 4 fail-safe contactors. Advisory on charging, protective on loads. ≈ 2.6 mA average draw.

Three validation gates — and what we need to pass the first.



What V-Guard gains

battery replacements at the right time · warranty by evidence · a premium 'intelligent' tier · a fleet-data moat

What we ask for

one opened Prime-series inverter + charger schematic · ESP32 bench parts · batteries and cyclers for Gate 1



Isolation

reinforced AMC1311 barrier; HV zone behind a moat; TVS on every line

Fail-safe

NC contactors — any fault, reset or brown-out brings loads back

Standards

IS 13252 / IEC 62368-1 (BIS gate) · IEC 62040 UPS · IEC 61000 EMC · IEEE 1159 / IEC 61000-4-30 Class-S-like PQ · UL94 V-0 enclosure

Charging

-24 mV/°C compensation, derate above 45 °C, stop at 58 °C, 2 s heartbeat revert to factory

**Coil off means loads on.
A window, never a date.
Offline first, fleet second.**

Thank you — we'd love your questions.
Team Codey Tingle (TI3271)

